

417522 — DATA MODELING AND VISUALIZATION

UNIT I — INTRODUCTION TO DATA MODELLING

Expanded In-Semester Examination Notes • Fourth Year AI & DS • SPPU 2020 Course

□ HOW TO USE THESE NOTES

Start with the basic meaning, then learn the formula/diagram, then memorize the highlighted exam points. The content follows the Unit-I syllabus and the uploaded lecture/reference notes. The syllabus lists Unit I as Basic Probability; Statistics, Parameter Estimation and Fitting a Distribution; Data Modeling Concepts; and an exit-poll sampling case study.

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UNIT I SYLLABUS	COVERED
Discrete & continuous random variables	✓
Independence & covariance	✓
Central Limit Theorem	✓
Chebyshev inequality	✓
Diverse discrete & continuous distributions	✓
Descriptive & graphical statistics	✓
Method of moments	✓
Maximum likelihood estimation	✓
Supertypes & subtypes	✓
Hierarchical data	✓
Recursive relationships	✓
Historical data	✓
Exit-poll sampling case study	✓

UNIT I — MASTER MEMORY MAP

□ REMEMBER

Probability → Random variables → Distributions → Statistics → Sampling → Estimation → Data modeling → Real-world sampling

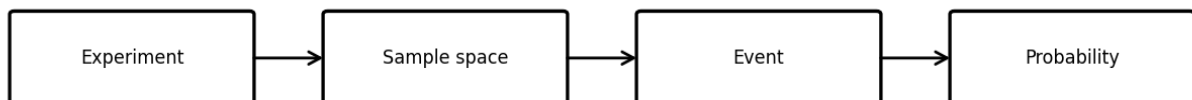
PART A — BASIC PROBABILITY

1. What is Probability?

Probability is a numerical measure of how likely an event is to occur. It is used to represent uncertainty. For an event A, the value lies between 0 and 1.

- $P(A)=0 \rightarrow$ event is impossible.
- $P(A)=1 \rightarrow$ event is certain.
- $0 < P(A) < 1 \rightarrow$ event has some chance of occurring.

Probability basics



Basic terms

- Experiment: an activity with an uncertain outcome, e.g., tossing a coin.
- Outcome: one possible result, e.g., Head.
- Sample space S: set of all possible outcomes.
- Event A: a collection of outcomes of interest.

Important probability rules

- Complement: $P(\bar{A}) = 1 - P(A)$.
- Addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Conditional probability: $P(A|B) = P(A \cap B) / P(B)$, when $P(B) > 0$.
- Multiplication rule: $P(A \cap B) = P(A)P(B|A)$.

Example: For a fair die, $P(4) = 1/6$. The event “even” contains $\{2, 4, 6\}$, so $P(\text{even}) = 3/6 = 1/2$.

EXAM POINT

For a 5-mark probability answer: definition $\rightarrow$ terms $\rightarrow$ 2-3 rules $\rightarrow$ small example.

2. RANDOM VARIABLE

A random variable assigns a numerical value to each outcome of a random experiment. It converts uncertain outcomes into numbers so that probability and statistics can be applied.

Example: Toss a coin three times and let X be the number of heads. Possible values are 0, 1, 2, 3.

3. DISCRETE RANDOM VARIABLE

A discrete random variable takes countable values. Usually the values arise from counting.

- Number of students in a class.
- Number of defective products.
- Number of phone calls in an hour.
- Number of accidents in a day.

MEMORY

DISCRETE = COUNTING

4. CONTINUOUS RANDOM VARIABLE

A continuous random variable can take any value within an interval. Values usually arise from measurement.

- Height, weight and temperature.
- Time taken to complete a task.
- Speed or distance.

□ MEMORY CONTINUOUS = MEASURING □		
Basis	Discrete	Continuous
Values	Countable	Any value in an interval
Typical source	Counting	Measurement
Distribution	PMF	PDF
Example	Number of calls	Waiting time

5. INDEPENDENCE

Two events are independent when occurrence of one event does not change the probability of the other. Mathematically, for independent A and B: $P(A \cap B) = P(A)P(B)$.

Example: Tossing a fair coin and rolling a die are independent when the two experiments do not affect one another.

Do not confuse independence with mutually exclusive events. Mutually exclusive events cannot occur together; independent events can occur together.

EXAM POINT

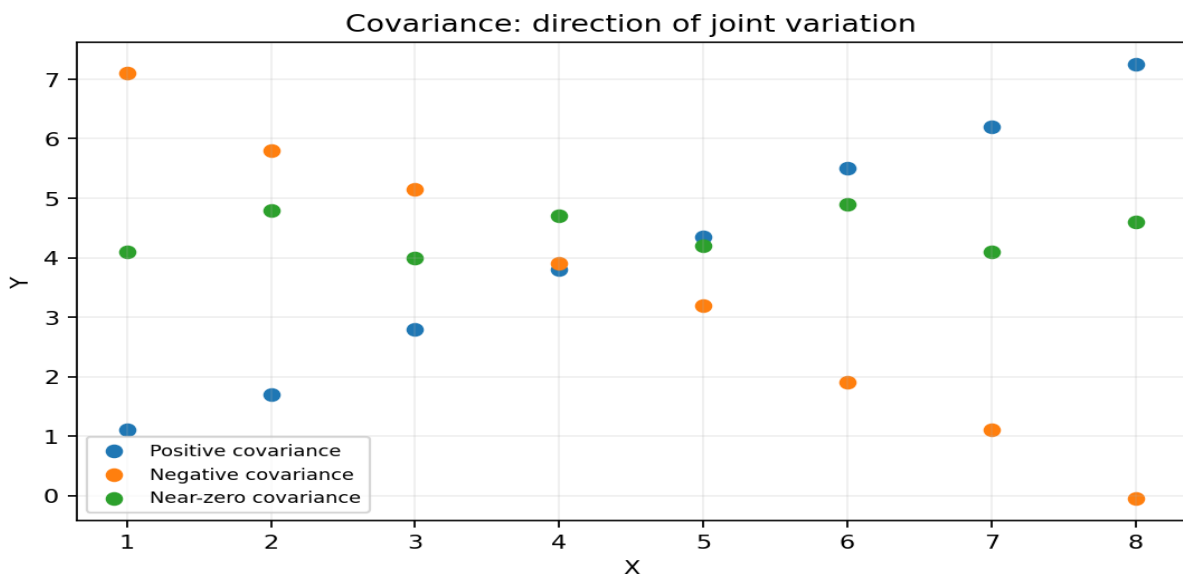
Write the formula $P(A \cap B) = P(A)P(B)$ and explain “one event does not influence the other.”

6. COVARIANCE

Covariance measures the direction of joint variation between two variables. It tells whether large values of one variable tend to occur with large values of the other, or with small values.

Population covariance: $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$.

Sample covariance: $s_{XY} = \sum[(X_i - \bar{X})(Y_i - \bar{Y})]/(n-1)$.



Type	Meaning	Simple example
Positive	Variables tend to increase/decrease together	Study hours ↑, marks ↑
Negative	One tends to increase while the other decreases	Price ↑, demand ↓
Near zero	Little linear co-movement	No clear linear pattern

Important interpretation

- Positive covariance does not prove causation.
- Covariance depends on measurement units, so its magnitude is not directly comparable across differently scaled variables.
- Covariance mainly communicates direction of linear joint variation.

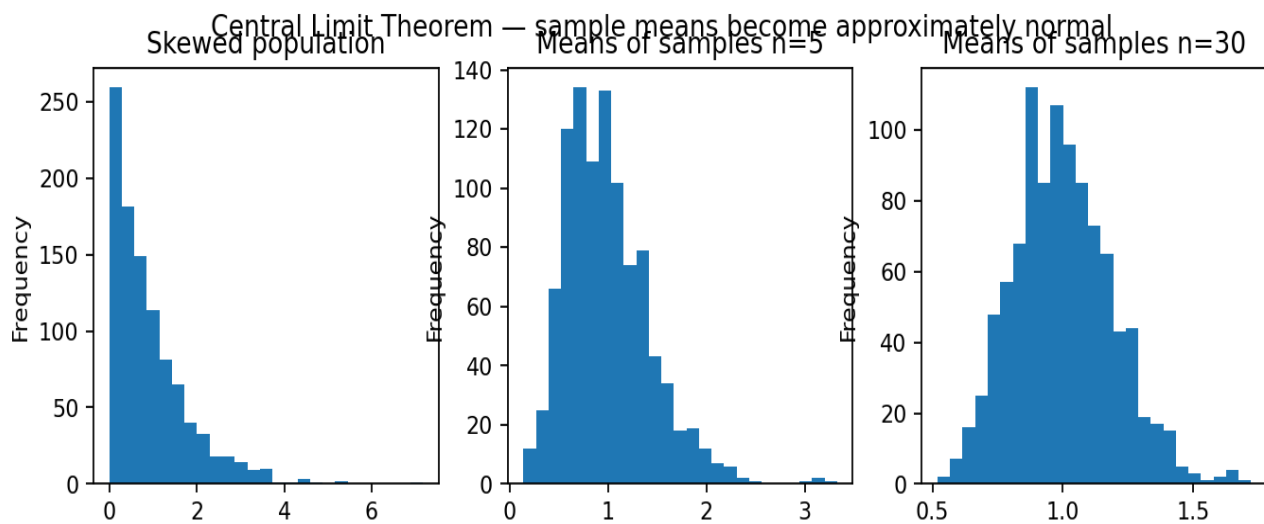
VERY IMPORTANT

Previous-year style question: “Explain positive, negative and zero covariance with appropriate graphs.” Learn the three meanings and the graph.

7. CENTRAL LIMIT THEOREM (CLT)

The Central Limit Theorem states that, under suitable conditions and for sufficiently large samples, the sampling distribution of the sample mean becomes approximately normal even when the original

population is not normal.



Key results

$$E(\bar{X}) = \mu$$
$$SE(\bar{X}) = \sigma / \sqrt{n}$$

- $\bar{X}$ is the sample mean.
- μ is the population mean.
- σ is the population standard deviation.
- n is sample size.
- Larger n generally makes the standard error smaller.

CLT — EASY EXAMPLE

Suppose delivery times in a city are right-skewed. A single delivery time may not follow a normal distribution. Now repeatedly select samples of, say, 30 deliveries and calculate the mean delivery time for every sample. The collection of those sample means becomes approximately bell-shaped when the usual CLT conditions are satisfied.

Why CLT matters

- It allows many statistical procedures about means to use normal-based reasoning.
- It connects sample information to population parameters.
- It explains why averages are often more stable than individual observations.

□ MEMORY

CLT = “INDIVIDUAL DATA CAN BE NON-NORMAL, BUT SAMPLE MEANS BECOME APPROXIMATELY NORMAL FOR LARGE ENOUGH n .”

8. CHEBYSHEV INEQUALITY

Chebyshev's inequality gives a guaranteed minimum proportion of observations lying within k standard deviations of the mean, without requiring a particular distribution shape.

$$P(|X - \mu| < k\sigma) \geq 1 - 1/k^2, k > 1$$

k	Minimum guaranteed proportion
2	75%
3	88.89%
4	93.75%

Example: At least 75% of observations lie within 2 standard deviations of the mean, regardless of the distribution, provided the conditions of the inequality are met.

□ REMEMBER

Chebyshev gives a GUARANTEE, not an exact percentage. Do not confuse it with the normal-distribution 68-95-99.7 rule.

9. DIVERSE DISCRETE AND CONTINUOUS DISTRIBUTIONS

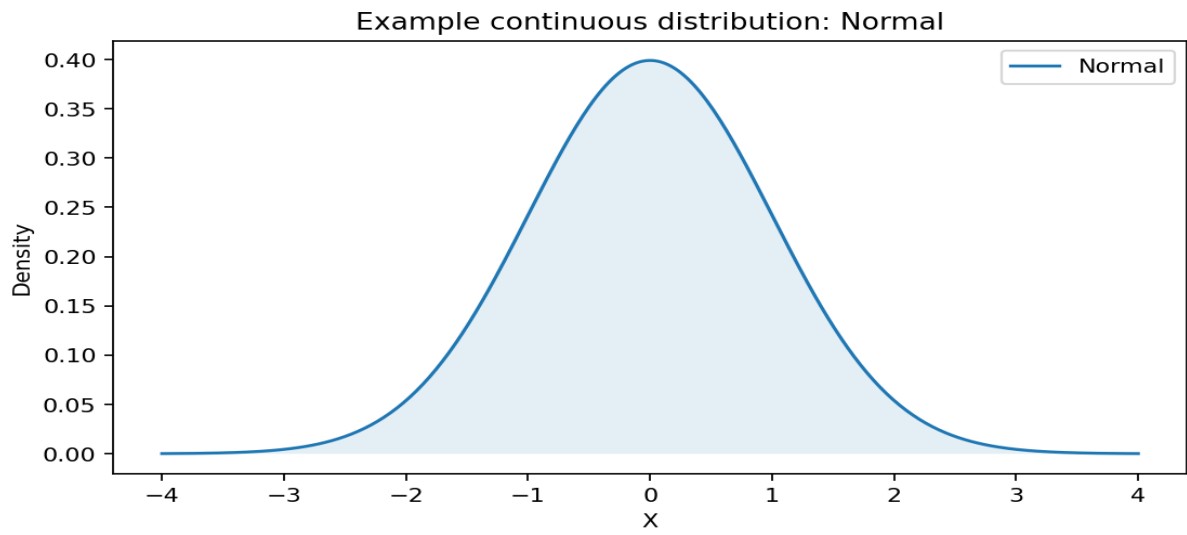
A probability distribution describes how probability is assigned to possible values of a random variable.

Discrete distributions

Distribution	Basic idea	Typical example
Bernoulli	One trial: success/failure	Pass/fail
Binomial	Number of successes in n trials	Number of defective items
Poisson	Number of events in an interval	Calls per hour

Continuous distributions

Distribution	Basic idea	Typical example
Uniform	Values spread evenly over an interval	Random time in a range
Normal	Symmetric bell-shaped distribution	Measurement errors
Exponential	Waiting/interarrival time model	Time between arrivals



10. STATISTICS — BASIC IDEA

Statistics is the process of collecting, organizing, summarizing, analyzing and interpreting data. In this unit, statistics helps us understand data and estimate unknown population characteristics.

Population vs sample

Term	Meaning	Example
Population	Entire group of interest	All students in a university
Sample	Selected part of population	500 selected students
Parameter	Numerical characteristic of population	μ , σ^2 , p
Statistic	Numerical characteristic of sample	$\bar{X}$, s^2 , $\hat{p}$

11. DESCRIPTIVE STATISTICS

Descriptive statistics summarizes observed data rather than making a claim about an unseen population.

Measures of central tendency

- Mean: arithmetic average; $\bar{X} = \sum X_i / n$.
- Median: middle value after sorting; useful when data contains extreme values.
- Mode: most frequently occurring value.

Measures of dispersion

- Range = maximum – minimum.
- Variance measures average squared deviation from the mean.
- Standard deviation = square root of variance and is expressed in the same units as the data.

Sample variance: $s^2 = \sum (X_i - \bar{X})^2 / (n - 1)$.

12. GRAPHICAL STATISTICS

Graphical statistics uses visual displays to make patterns, spread, comparisons, relationships and unusual observations easier to see.

Graph	Best use
Bar chart	Compare categories
Histogram	Show distribution of numerical values
Pie chart	Show parts of a whole
Box plot	Show median, quartiles, spread and possible outliers
Scatter plot	Study relationship between two numerical variables

□ MEMORY

BAR = categories | HISTOGRAM = distribution | BOX = spread/outliers | SCATTER = relationship

13. PARAMETER ESTIMATION

A population parameter is usually unknown. We use sample data to estimate it. For example, sample mean $\bar{X}$ can estimate population mean μ .

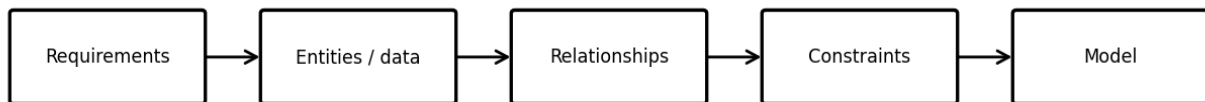
Point estimation gives one numerical estimate. Example: if $\bar{X} = 72$, then 72 is a point estimate of μ .

14. FITTING A DISTRIBUTION

Distribution fitting means finding a probability distribution that reasonably represents observed data. The aim is not to force every dataset into a distribution; the selected model should be appropriate for the data and its purpose.

General fitting idea

Data modeling



- Collect/inspect observed data.
- Identify plausible distribution family from data characteristics and subject knowledge.
- Estimate the distribution parameters from the sample.
- Compare the fitted model with the observed data.
- Use the model only if the fit is reasonably appropriate.

□ MEMORY

DATA → CHOOSE DISTRIBUTION → ESTIMATE PARAMETERS → CHECK FIT → USE MODEL

15. METHOD OF MOMENTS (MoM)

Method of Moments estimates unknown parameters by equating sample moments with the corresponding theoretical moments.

Easy meaning: the sample has measurable characteristics such as an average. The theoretical distribution has expressions for those characteristics. We make them equal and solve for the unknown parameter(s).

Steps

- Write the first few sample moments.
- Write the corresponding theoretical moments of the chosen distribution.
- Equate sample moments to theoretical moments.
- Solve the equations for the unknown parameters.

Simple idea example: If a distribution has theoretical mean equal to θ and the sample mean is $\bar{X}$, then the first-moment estimate is $\hat{\theta} = \bar{X}$.

□ EXAM POINT

MoM keyword = MATCH sample moments with theoretical moments.

16. MAXIMUM LIKELIHOOD ESTIMATION (MLE)

MLE chooses parameter values that make the observed sample most likely under the assumed probability model.

Likelihood: $L(\theta) = P(\text{observed data} \mid \theta)$. The MLE is the parameter value θ that maximizes $L(\theta)$.

Steps

- Specify the probability model.
- Write the likelihood function for the observed sample.
- Often take the logarithm to simplify multiplication into addition.
- Differentiate with respect to the parameter.
- Set derivative to zero and solve for the candidate estimate.
- Check that the candidate gives a maximum.

Example — Bernoulli coin: If a coin is tossed n times and x heads are observed, the likelihood is proportional to $p^x(1-p)^{n-x}$. Maximizing it gives $\hat{p}=x/n$.

17. METHOD OF MOMENTS vs MLE

Point	Method of Moments	Maximum Likelihood
Core idea	Match sample and theoretical moments	Maximize likelihood of observed data
Main object	Moments	Likelihood function
Typical approach	Solve moment equations	Optimization/calculus often used
Memory	MATCH	MAXIMIZE

❏ VERY IMPORTANT

Previous-year style questions frequently ask MoM, MLE, or their difference. Learn the definitions, steps and one small example.

PART C — DATA MODELING CONCEPTS

18. DATA MODEL — BASIC UNDERSTANDING

A data model is an abstract representation of data. It describes what data exists, what the data means, how data objects are related and what rules/constraints apply.

Example: a college may have STUDENT, TEACHER, SUBJECT and DEPARTMENT. Relationships can represent which student studies which subject or which teacher teaches which subject.

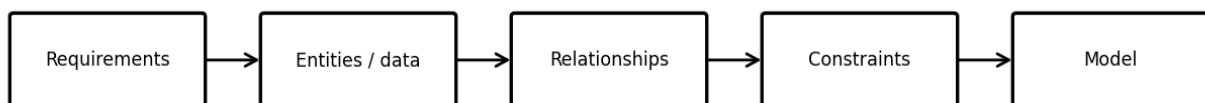
Why model data?

- Organizes information clearly.
- Represents relationships between objects.
- Reduces ambiguity about meaning.
- Supports consistency and useful constraints.
- Makes data easier to store and analyze.

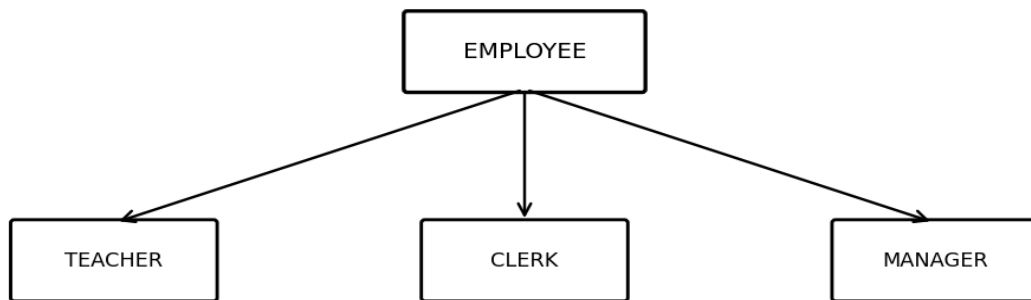
19. SUPERTYPES AND SUBTYPES

A supertype is a general entity class containing common characteristics. A subtype is a specialized category of that supertype.

Data modeling



Supertype → Subtypes



Example: EMPLOYEE can contain Employee ID and Name. TEACHER can have specialization-related information; CLERK and MANAGER can have their own specific attributes.

- Benefits: reduces repetition.
- Represents real-world classification.
- Allows common attributes to be stored at the general level.
- Keeps specialized attributes with the relevant subtype.

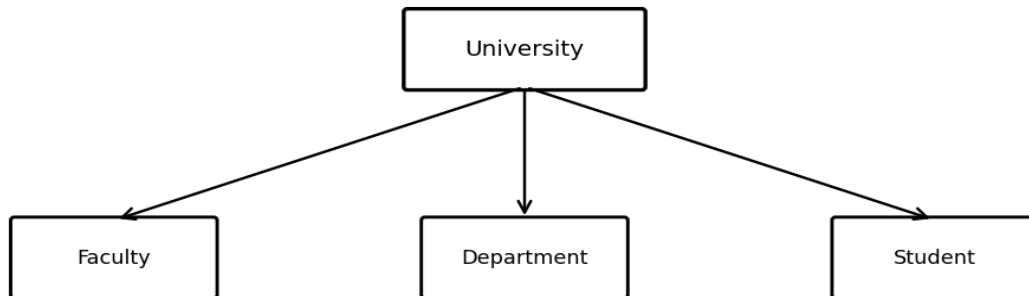
□ MEMORY

SUPERTYPE = GENERAL □ | SUBTYPE = SPECIFIC □

20. HIERARCHICAL DATA

Hierarchical data is organized in a tree-like structure using parent-child relationships. A higher-level object can have one or more lower-level child objects.

Hierarchical structure



- Root: top-level node.
- Parent: node that has child nodes.
- Child: node below a parent.
- Path: route from one node through connected parent-child levels.

Example: University → Faculty → Department → Student.

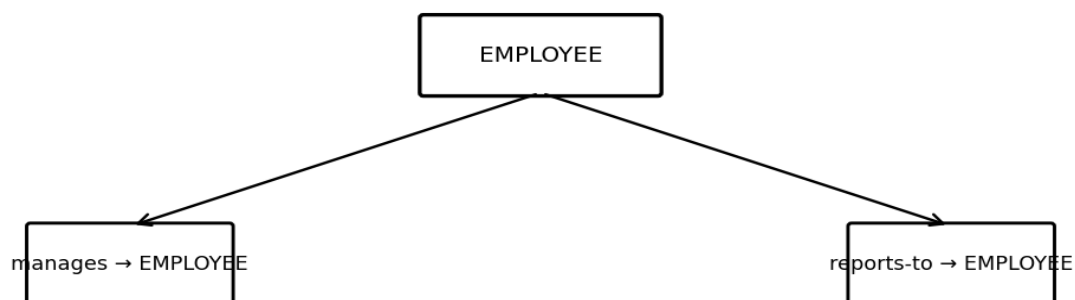
□ MEMORY

HIERARCHY = TREE = PARENT → CHILD

21. RECURSIVE RELATIONSHIP

A recursive relationship occurs when an entity has a relationship with another instance of the same entity. It is also called a self-referencing relationship.

Recursive relationship

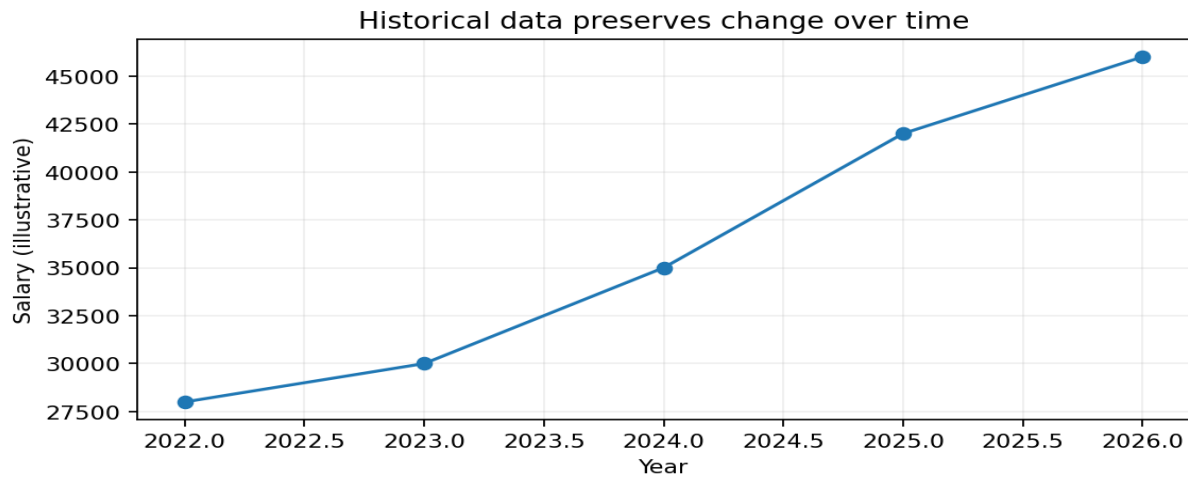


Example: EMPLOYEE manages EMPLOYEE. Both objects are employees, but one has the role of manager and the other has the role of subordinate.

Another simple example: PERSON is parent-of PERSON.

22. HISTORICAL DATA

Historical data stores past states, past events or previous values. It is important when the current value alone cannot explain how the system changed.



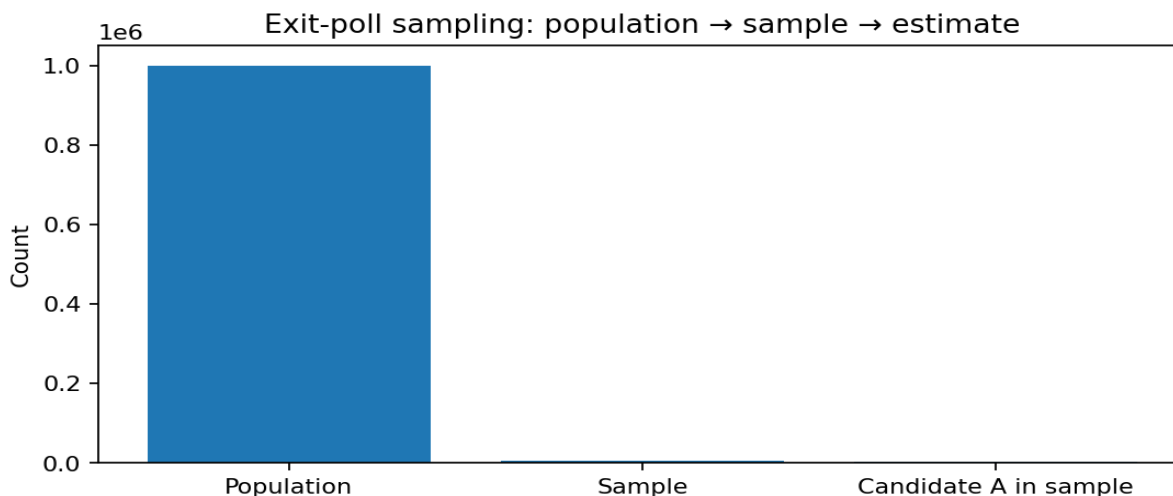
- Supports trend analysis.
- Supports comparisons across time.
- Supports auditing and traceability.
- Helps understand past behavior and changes.

Example: An employee salary may be ₹30,000 in 2023, ₹35,000 in 2024 and ₹42,000 in 2025. Storing only ₹42,000 loses the historical information.

Historical data can be modeled using historical records, event histories or current-plus-history structures, depending on the requirement.

23. EXIT-POLL SAMPLING — SYLLABUS CASE STUDY

An exit poll collects information from a selected sample of voters after they leave polling stations. The sample is used to estimate characteristics of the larger voter population.



Example

Suppose the population contains 10,00,000 voters and 5,000 are sampled. If 2,300 sampled voters report Candidate A, then the sample proportion is $\hat{p} = 2300/5000 = 0.46$, or 46%.

Complete process

- Define the target population.
- Choose a sample that aims to represent the population.
- Collect responses from selected voters.
- Calculate sample statistics, such as proportions.
- Use the sample statistic to estimate the population characteristic.
- Report the result with appropriate uncertainty/caution.

Why sampling is used

- A complete population survey can be expensive and time-consuming.
- A well-designed sample makes large-population studies practical.
- Sampling produces an estimate rather than a guaranteed exact population value.

Main sources of error/bias

- Sampling bias: selected sample is not representative.
- Non-response: selected people do not provide responses.
- Undercoverage: some population members have little/no chance of selection.
- Question wording can influence responses.
- Sampling variability: different samples can produce different estimates.

EXAM POINT

This case study is explicitly in the syllabus. Draw/describe: Population → Sample → Responses → Statistic → Population estimate.

UNIT I — EXAM PRIORITY

Very High	CLT with example + diagram
Very High	Positive, negative and zero covariance with graph
Very High	MLE with steps + example

☐ Very High	CLT with example + diagram
☐ Very High	MoM + MoM vs MLE
☐ Very High	Supertype/subtype with diagram
☐ High	Discrete vs continuous random variables
☐ High	Chebyshev inequality
☐ High	Discrete/continuous distributions
☐ High	Descriptive vs graphical statistics
☐ High	Hierarchical, recursive, historical data
☐ High	Exit-poll sampling case study

UNIT I — FINAL MEMORY CHART ☐

Topic	Remember in one line
Probability	Likelihood of an event
Discrete RV	COUNT
Continuous RV	MEASURE
Independence	No probability influence
Covariance	Direction of joint variation
CLT	Large-sample means $\approx$ normal under suitable conditions
Chebyshev	Minimum guaranteed proportion within $k\sigma$
Distribution	Probability pattern for a random variable
Descriptive statistics	Summarize observed data
Graphical statistics	See patterns visually
Parameter estimation	Use sample to estimate population parameter
Fitting	Choose + estimate + check distribution
MoM	MATCH moments
MLE	MAXIMIZE likelihood
Supertype	General
Subtype	Specific
Hierarchy	Parent $\rightarrow$ child
Recursive	Same entity $\rightarrow$ itself
Historical	Past states/events
Exit poll	Sample voters $\rightarrow$ estimate population

5-MARK ANSWER FORMULA

Definition $\rightarrow$ key explanation $\rightarrow$ formula/steps $\rightarrow$ suitable example $\rightarrow$ diagram/table $\rightarrow$ final key point. Do not write only a definition for a 5-mark question.

☐ SYLLABUS CONTROL

No separate advanced topic has been intentionally added as an examination requirement. Extra explanations here are basic supporting explanations needed to understand the listed syllabus points. The official syllabus lists the Unit-I boundary. filecite $\rightarrow$ turn25file7 $\rightarrow$